Chapter 3 Assessment

Math 90-S61, Fall 2026, Jack Love

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Last name, first name, ID#: SOLUTIONS

Please read the following instructions carefully.

- You have until 12:25 PM to complete the assessment.
- Use the backs of pages for rough work/scratch paper. Show all relevant work neatly in the spaces provided, and box all final answers.
- You are welcome to ask me clarifying questions.

1. (4 pts) Suppose $Q = 27.03(1.67)^t$. Give the starting value a, the growth factor b, and the growth rate r. In other words, determine a, b, and r such that $Q = ab^t = a(1+r)^t$.

$a = 27.03$ $\quad b = 1.67$ $\quad r = 0.67$

2. (8 pts) Suppose the annual inflation rate is 4.1% per year.

(a) (4 pts) If a movie ticket costs \$12.75 today, find a formula for $p(t)$, the cost of a movie ticket t years from today, assuming that movie tickets keep up with inflation.

$p(t) = 12.75(1.041)^t$

(b) (4 pts) According to your formula, how much will a movie ticket cost in 25 years? Round your answer to the nearest cent.

$p(25) = 12.75(1.041)^{25} \approx \$34.82$

3. (8 pts) A population has size 6500 at time $t = 0$, with t in years.

(a) (4 pts) If the population decreases by 60 people per year, find a formula for $P(t)$, the population at time t.

$P(t) = 6500 - 60t$

(b) (4 pts) If the population decreases by 5% per year, find a formula for $P(t)$, the population at time t.

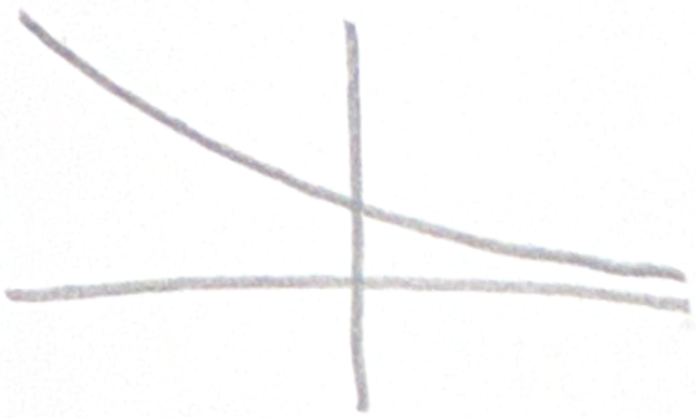
$P(t) = 6500(0.95)^t$

4. (4 pts) If $0 < b < 1$, what is the equation of the horizontal asymptote of $y = ab^t$ as $t \to \infty$? If there is no horizontal asymptote, say so.

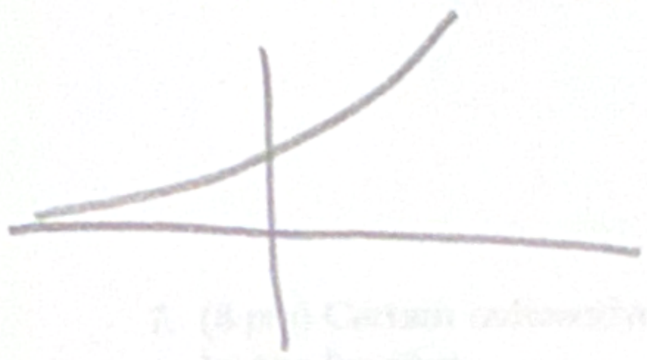
$y = 0$

5. (16 pts) Find the long term behavior of each of the following functions. Show graphs or algebraic work to support your answers.

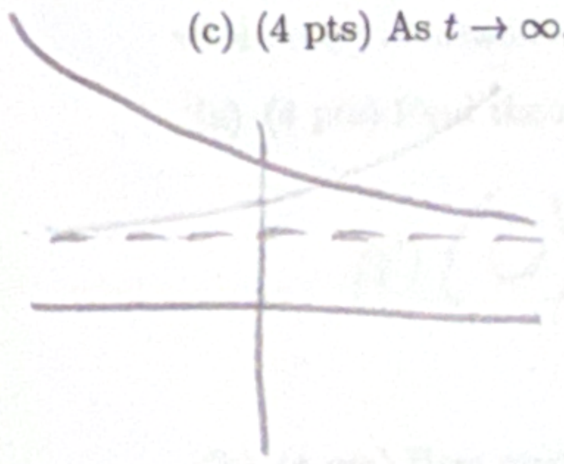
(a) (4 pts) As $x \to \infty$, $192(0.75)^x \to$ 0



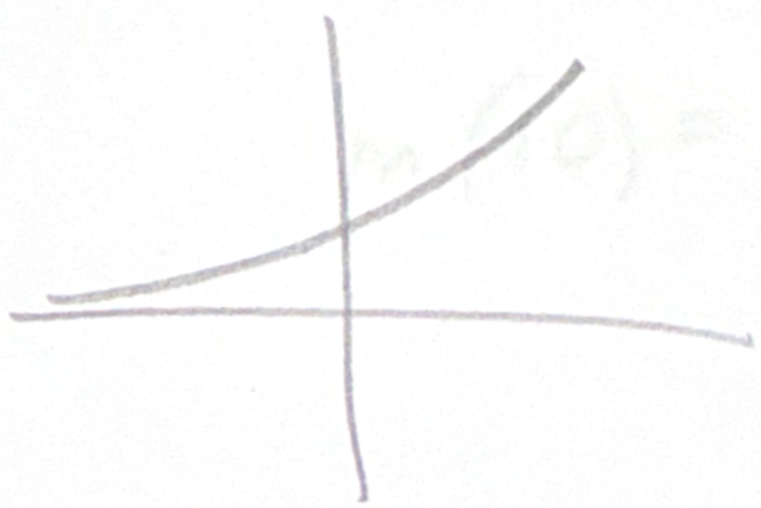
(b) (4 pts) As $t \to -\infty$, $2(3)^t \to$ 0



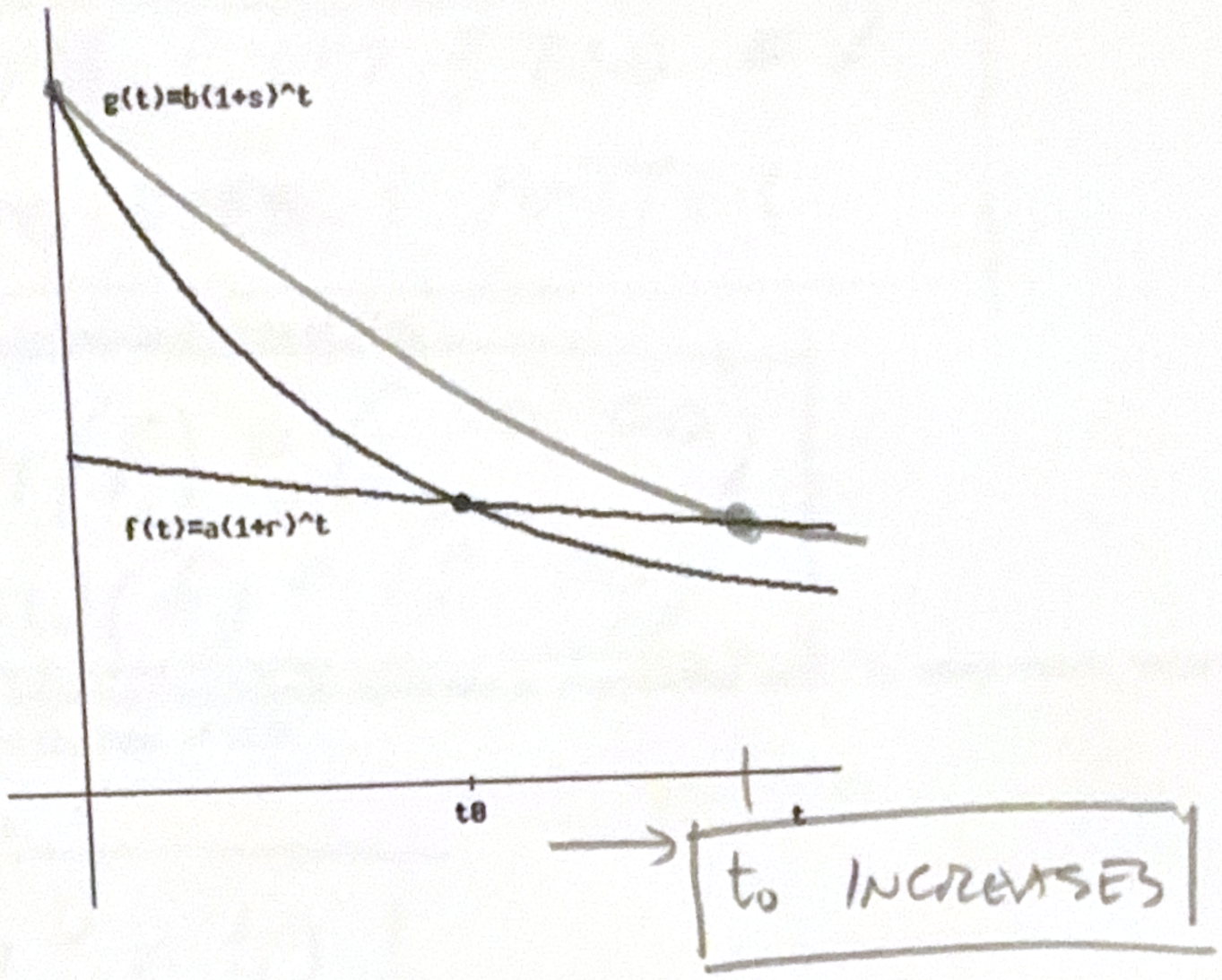
(c) (4 pts) As $t \to \infty$, $5(3 + (0.9)^t) \to$ 15



(d) (4 pts) As $t \to -\infty$, $8e^{0.5t} \to$ 0



6. (4 pts) Suppose t_0 is the point of intersection of the graphs below, where $f(t) = a(1 + r)^t$ and $g(t) = b(1 + s)^t$. If the value of s is increased and the values of a, b, and r remain the same, does the value of t_0 increase, decrease or stay the same? Sketch the graph of the new $g(t)$ with increased s to support your answer.



7. (8 pts) Certain radioactive material decays in such a way that the mass remaining after t years is given by the function

$$m(t) = 160e^{-0.025t}$$

where $m(t)$ is measured in grams.

(a) (4 pts) Find the mass at time $t = 0$.

$m(0) = 160e^{-0.025 \cdot 0} = \boxed{160}$

(b) (4 pts) How much of the mass remains after 10 years? (State the exact answer in terms of e, or approximate your answer to 4 decimal places.

$m(10) = 160e^{-0.025 \cdot 10}$

$= \boxed{160e^{-0.25}}$

$= \boxed{\frac{160}{e^{0.25}} = \frac{160}{\sqrt[4]{e}}}$

$\approx \boxed{124.6081}$

8. (8 pts) The graph of the function $f(x) = e^{x-4} + 2$ can be obtained from the graph of $g(x) = e^x$ by performing two transformations.

(a) (4 pts) What are the transformations? Does the order in which you perform the transformations matter? If so, state the correct order.

TRANSLATE RIGHT 4 AND UP 2.

ORDER DOESN'T MATTER.

(b) (4 pts) State the domain and range of $f(x)$.

$DOM(f) = (-\infty, \infty)$

$RAN(f) = (2, \infty)$

9. (8 pts) Express the following logarithmic equations in exponential form. In other words, write an equivalent equation of the form $e^A = B$.

(a) (4 pts) $\ln 10 = y$

$e^y = 10$

(b) (4 pts) $\ln x = 5$

$e^5 = x$

10. (12 pts) Evaluate the following expressions. Your answers must be exact and in simplest form.

(a) (4 pts) $\ln(e^{1.7})$

1.7

(b) (4 pts) $e^{\ln 4}$

4

(c) (4 pts) $\ln\left(\frac{1}{e^3}\right)$

$\ln\left(\frac{1}{e^3}\right) = \ln(e^{-3}) = -3$

11. (16 pts) Find the exact solutions of the following equations.

(a) (4 pts) $e^{-3x+2} = 10$.

$e^{-3x+2} = 10 \Rightarrow -3x+2 = \ln(10) \Rightarrow$

$-3x = \ln(10) - 2 \Rightarrow \boxed{x = -\frac{1}{3}(\ln(10) - 2)}$

(b) (4 pts) $8 - \ln(2 - x) = 1$.

$8 - \ln(2-x) = 1 \Rightarrow -\ln(2-x) = -7 \Rightarrow$

$\ln(2-x) = 7 \Rightarrow 2 - x = e^7 \Rightarrow -x = e^7 - 2 \Rightarrow$

$\boxed{x = 2 - e^7}$

(c) (4 pts) $4^x = 3$

$4^x = 3 \Rightarrow \ln(4^x) = \ln(3) \Rightarrow$

$x\ln(4) = \ln(3) \Rightarrow \boxed{x = \frac{\ln(3)}{\ln(4)}}$

(d) (4 pts) $2 \cdot 4^{2x-4} = 60$

$2 \cdot 4^{2x-4} = 60 \Rightarrow 4^{2x-4} = 30 \Rightarrow$

$\ln(4^{2x-4}) = \ln(30) \Rightarrow (2x-4)\cdot\ln(4) = \ln(30)$

$\Rightarrow 2x - 4 = \frac{\ln(30)}{\ln(4)} \Rightarrow 2x = \frac{\ln(30)}{\ln(4)} + 4 \Rightarrow$

$\boxed{x = \frac{1}{2}\cdot\frac{\ln(30)}{\ln(4)} + 2}$

12. (4 pts) The temperature of a cup of coffee obeys Newton's law of cooling. Suppose the coffee has a temperature of 200 degrees Fahrenheit when freshly poured, and 2 minutes later has cooled to 190 degrees in a room at 75 degrees. Determine a formula $F(t) = ae^{-kt} + c$ for the temperature of the coffee t minutes after it was poured.

$F(0) = 200$. $F(0) = ae^{-k\cdot 0} + c = a + c$

$\boxed{a + c = 200}$

$F \to 75$ as $t \to \infty$. $F \to c$ as $t \to \infty$.

$\boxed{c = 75}$ $\boxed{a = 125}$

$F(t) = 125e^{-kt} + 75$.

$F(2) = 190$. $F(2) = 125e^{-2k} + 75$.

$190 = 125e^{-2k} + 75 \Longrightarrow 115 = 125e^{-2k} \Longrightarrow$

$e^{-2k} = \frac{115}{125} \Longrightarrow -2k = \ln\left(\frac{115}{125}\right) \Longrightarrow$

$k = -\frac{1}{2} \cdot \ln\left(\frac{115}{125}\right)$.

$\boxed{F(t) = 125e^{\frac{1}{2}\ln\left(\frac{115}{125}\right)t} + 75}$